

DECAL COLLIDER USAGE GUIDE

Unity Decal Projection & Automatic MeshCollider Generation

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1. Introduction and Overview

Decal Collider is a system designed to project a decal (visual) onto surfaces within a projection volume you define in the scene, while simultaneously generating a MeshCollider suitable for this decal. It supports both fast iteration within the Editor and runtime automation (API) scenarios.

1.1 Why Decal Collider?

- Single system, single source: Decal visual + collider generation are managed in the same component.
 - Controlled projection: Wraps only around desired surfaces using LayerMask (Wrap Mask).
 - Production quality: You can adjust the visual mesh and collider mesh density separately.
 - Automation ready: Integrates into the pipeline with async rebuild, hit callbacks, and helper functions.
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2. System Requirements and Installation

2.1 Requirements

- Unity: Unity 2021.3 LTS or higher is recommended.
- Platform: Windows / macOS / Linux (Editor + Runtime).
- Render Pipeline: Built-in / URP / HDRP (pipeline-agnostic).

2.2 Importing the Package

1. Download the package you purchased from the Unity Asset Store.
2. In Unity, open the Assets > Import Package > Custom Package... menu.
3. Ensure all files are selected and press Import.

2.3 Shader Graph Dependency

Shader Graph may be required for some visual workflows; if Shader Graph is not installed, you may see a warning/installation prompt when opening the package. Installing Shader Graph via the Package Manager and importing it again will make the visual side seamless.

3. Getting Started (Quick Setup)

1. Select a GameObject in the scene (an empty object or one specific to the decal).
2. Add the Add Component > Physics > Decal Collider component.

3. If a MeshCollider is not added automatically, add a MeshCollider to the same object (mandatory).
4. Select the projection mode (GridProjection or MeshProjection) and trigger the generation with Refresh/Rebuild.

Note: If using MeshProjection, inputMesh can be automatically generated via the SpriteRenderer or TextMeshPro source; in GridProjection mode, it is recommended to generate a "plane" using MeshFilter/MeshRenderer.

4. Projection and Collider Generation

4.1 GridProjection (Plane) Flow

GridProjection creates a grid in the Size area you specify and calculates the intersection with the surface by casting a raycast in the chosen projection direction. This mode is fast for ground decals (trails, stains, etc.).

1. Decal Mode: Select GridProjection.
2. Define the layers for projection using Wrap Mask.
3. Adjust the Size (X/Y) and Max Distance values based on the area the decal will cover in the scene. Adjust the Mesh Subdivisions (visual) and Collider Subdivisions (physics) values according to your needs.
4. Generate the mesh and collider using Refresh/Rebuild.

4.2 MeshProjection (Sprite / TMP)

MeshProjection uses a Mesh, Sprite, or TextMeshPro output as a projection source.

- With Sprite: If the object has a SpriteRenderer, the system generates an inputMesh based on the sprite bounds and synchronizes the size/center.
- With TMP: A snapshot of the TextMeshPro mesh is taken and used as the projection source; Custom Mesh: You can assign a Mesh directly to the inputMesh field.

4.3 Rebuild / Debug Tools

- Refresh: Triggers a safe rebuild after parameter changes (RebuildSafe logic).
 - Edit Collider / Edit Projection: Allows you to adjust Max Distance and direction/rotation using scene view handles.
 - Bounds Tool / Debug Rays: Visualizes the raycast area; shows hit points with debug lines.
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5. Smart Settings and Controls

5.1 Wrap Mask, Size, Max Distance

- Wrap Mask (LayerMask): Limits the surfaces the raycast will hit (performance + control).
- Size: Projection area (X/Y).
- Max Distance: Projection depth (Z). If too low, it won't reach the surface; if too high, it collects unnecessary hits.
- Center: Shifts the projection volume with a local offset.

5.2 Subdivisions, Dynamic LOD, Culling

- Mesh Subdivisions: Visual mesh resolution. Higher values give smoother results, but increase cost.
- Collider Subdivisions: Collider resolution. Generally kept lower than the visual resolution.
- Use Dynamic LOD: Automatically reduces subdiv values based on distance to the camera in Play mode.
- Cull If Invisible: Preserves performance by deferring rebuilds if the camera is not looking.
- Ignore Self: Reduces the risk of hitting its own collider (self-hit).

5.3 Alpha Masking (Clipping by Transparency)

Alpha Masking filters collider triangles based on the transparency in the decal's texture; it is ideal for reducing unnecessary collider area, especially in round/organic decals.

- Alpha Threshold: Regions below the threshold value are clipped from the collider.
- GridProjection note: The material's mainTexture (Texture2D) must be readable (Read/Write) for the mask texture.

6. Advanced Features and Automation

6.1 Live Link (TMP / Sprite Updates)

In MeshProjection mode, if TMP text or Sprite changes, you can check for changes at specific intervals and rebuild using Live Link.

- Update Interval: How many seconds between checks.
- Update Collider: If enabled, the collider is also updated (more costly). If disabled, only the visual mesh is updated.

6.2 Scripting API - Quick Examples

Basic functions for runtime automation scenarios via DecalCollider.API:

- ForceRebuild: Starts a synchronous or asynchronous rebuild.
- AsyncRebuild: Rebuilds with a progress callback via IEnumerator.

- **SetSprite / SetText:** Changes the source content and quickly regenerates.
- **RegisterOnHitCallback:** Binds a filter + callback for raycast hits.
- **DebugDrawHitPoints:** Draws hit points in the scene (debug).

```
// Rebuild (sync) veya async
decal.ForceRebuild(true);
StartCoroutine(decal.AsyncRebuild(p => Debug.Log($"Progress: {p:0.00}")));

// Sprite / TMP güncelleme
decal.SetSprite(newSprite);
decal.SetText("HELLO");

// Hit callback
decal.RegisterOnHitCallback(h => h.collider != null,
    h => Debug.Log(h.collider.name));
```

6.3 Saving Mesh as an Asset

You can use `SaveMeshToAsset()` to save the visual mesh generated within the Editor as an asset file. This is useful for packaging or versioning static decal meshes.

7. Troubleshooting & Performance

7.1 Common Issues

- **Decal does not sit on the surface:** Check the Wrap Mask layer; increase Max Distance; verify the projection direction.
- **Collider does not generate:** Ensure there is a MeshCollider on the object; run Refresh/Rebuild.
- **Alpha masking is not working:** Is Texture2D Read/Write enabled? Is mainTexture assigned in GridProjection?
- **Self-hit / distortion:** Enable Ignore Self or move the projection volume slightly away from the collider.

7.2 Performance Tips

- **Keep collider subdiv low:** Physics cost is directly affected by the tri count.
- **Cull If Invisible:** Makes a big difference if there are many decals.
- **Dynamic LOD:** Provides stable FPS by automatically reducing quality for distant decals.
- **Update Collider disabled in Live Link:** Recommended for frequently updated TMPs/Sprites.

8. Support and Feedback

Documentation: Visit our [documentation](#) site for more information and usage examples:

- Community: Join our [Discord](#) community for help, tips, and to share your feedback.
- Review: If this tool has accelerated your workflow, please leave a review on the [Asset Store](#).